

SDS8102: Advanced Mathematical Statistics

Lecture 8: Fisher Information, Efficiency, and Local Geometry

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Lecture 8: Information and Efficiency

Suppose we want to estimate

$$g(\theta).$$

Last lecture asked how to compare estimators using their risk.

Today we ask a different question:

How accurately can $g(\theta)$ possibly be estimated?

We will develop

unbiasedness $\rightarrow$ Fisher information $\rightarrow$ Cramér–Rao bound $\rightarrow$ efficiency.

We will then connect Fisher information to exponential families and the local geometry of KL divergence.

Unbiased Estimation

An estimator $T(X)$ of $g(\theta)$ is **unbiased** if

$$\mathbb{E}_\theta[T(X)] = g(\theta) \quad \forall \theta.$$

Under squared error loss,

$$R(\theta, T) = \text{Var}_\theta(T) + (\mathbb{E}_\theta[T] - g(\theta))^2.$$

Therefore, among unbiased estimators,

$$R(\theta, T) = \text{Var}_\theta(T).$$

This suggests asking:

How small can the variance of an unbiased estimator be?

Not Every Function Has an Unbiased Estimator

Suppose

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta).$$

For any statistic $T(X_1, \dots, X_n)$,

$$\mathbb{E}_\theta[T] = \sum_{x \in \{0,1\}^n} T(x) \theta^{\sum_i x_i} (1 - \theta)^{n - \sum_i x_i}.$$

Hence

$$\mathbb{E}_\theta[T] \text{ is a polynomial in } \theta \text{ of degree at most } n.$$

Therefore there cannot exist an unbiased estimator of, for example,

$$\frac{\theta}{1 - \theta}.$$

Unbiasedness itself can impose strong structural restrictions.

The Score Function

Let X have density $f_\theta(x)$.

The **score function** is

$$S_\theta(X) = \frac{\partial}{\partial \theta} \log f_\theta(X).$$

Equivalently,

$$S_\theta(X) = \frac{\partial_\theta f_\theta(X)}{f_\theta(X)}.$$

Under suitable regularity conditions,

$$\mathbb{E}_\theta[S_\theta(X)] = \int \partial_\theta f_\theta(x) dx = \frac{\partial}{\partial \theta} \int f_\theta(x) dx = 0.$$

Thus

$$\mathbb{E}_\theta[S_\theta(X)] = 0.$$

Since the score has mean zero, its variance measures its typical magnitude.

Definition: Fisher Information

The Fisher information in X about θ is

$$I_X(\theta) = \mathbb{E}_\theta [S_\theta(X)^2] .$$

Intuitively, if a small change in θ produces a large change in the likelihood, the data contain more information about θ .

Large $I_X(\theta)$ means that nearby parameter values are easier to distinguish.

An Equivalent Formula

Recall

$$S_\theta(X) = \partial_\theta \log f_\theta(X).$$

Differentiate again:

$$\partial_\theta^2 \log f_\theta = \frac{\partial_\theta^2 f_\theta}{f_\theta} - \left(\frac{\partial_\theta f_\theta}{f_\theta} \right)^2.$$

Taking expectation,

$$\mathbb{E}_\theta[\partial_\theta^2 \log f_\theta(X)] = \int \partial_\theta^2 f_\theta(x) dx - I_X(\theta).$$

Under regularity conditions, the first term is zero. Hence

$$I_X(\theta) = -\mathbb{E}_\theta \left[\frac{\partial^2}{\partial \theta^2} \log f_\theta(X) \right].$$

Example: Gaussian Location

Suppose

$$X \sim N(\theta, \sigma^2).$$

The log-likelihood is

$$\ell(\theta; X) = -\frac{(X - \theta)^2}{2\sigma^2} + \text{constant}.$$

Therefore

$$S_{\theta}(X) = \frac{X - \theta}{\sigma^2}.$$

Hence

$$I_X(\theta) = \mathbb{E}_{\theta} \left[\frac{(X - \theta)^2}{\sigma^4} \right] = \boxed{\frac{1}{\sigma^2}}.$$

More noise means less information.

Information Adds Across Independent Observations

Suppose

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f_\theta.$$

The joint score is

$$S_n(\theta) = \sum_{i=1}^n S_\theta(X_i).$$

Since the individual scores are independent and have mean zero,

$$I_n(\theta) = \text{Var}_\theta \left(\sum_{i=1}^n S_\theta(X_i) \right) = n I_1(\theta).$$

Thus

$$I_n(\theta) = n I_1(\theta).$$

Information grows linearly with sample size.

This will explain why estimation error typically scales as $1/\sqrt{n}$.

The Key Identity

Suppose $T(X)$ is unbiased for $g(\theta)$:

$$\mathbb{E}_\theta[T(X)] = g(\theta).$$

Differentiate both sides:

$$g'(\theta) = \frac{\partial}{\partial \theta} \int T(x) f_\theta(x) dx.$$

Under regularity conditions,

$$g'(\theta) = \int T(x) \partial_\theta f_\theta(x) dx.$$

Since

$$\partial_\theta f_\theta(x) = f_\theta(x) S_\theta(x),$$

we obtain

$$g'(\theta) = \mathbb{E}_\theta[T(X) S_\theta(X)].$$

Covariance with the Score

Recall

$$\mathbb{E}_\theta[S_\theta(X)] = 0.$$

Therefore

$$\mathbb{E}_\theta[T(X)S_\theta(X)] = \text{Cov}_\theta(T(X), S_\theta(X)).$$

Hence the key identity is

$$\boxed{\text{Cov}_\theta(T, S_\theta) = g'(\theta).}$$

This identity connects

$\boxed{\text{estimation of } g(\theta) \quad \text{to} \quad \text{sensitivity of the likelihood to } \theta.}$

Cauchy–Schwarz now gives a fundamental lower bound.

Cramér–Rao Lower Bound

By Cauchy–Schwarz,

$$\text{Cov}_\theta(T, S_\theta)^2 \leq \text{Var}_\theta(T) \text{Var}_\theta(S_\theta).$$

Using

$$\text{Cov}_\theta(T, S_\theta) = g'(\theta)$$

and

$$\text{Var}_\theta(S_\theta) = I(\theta),$$

we obtain

$$[g'(\theta)]^2 \leq \text{Var}_\theta(T) I(\theta).$$

Therefore

$$\boxed{\text{Var}_\theta(T) \geq \frac{[g'(\theta)]^2}{I(\theta)}}.$$

This is the **Cramér–Rao lower bound**.

Cramér–Rao for an IID Sample

For

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f_\theta,$$

we have

$$I_n(\theta) = nl_1(\theta).$$

Hence every unbiased estimator T of $g(\theta)$ satisfies

$$\text{Var}_\theta(T) \geq \frac{[g'(\theta)]^2}{nl_1(\theta)}.$$

In particular, for unbiased estimation of θ itself,

$$g(\theta) = \theta, \quad g'(\theta) = 1,$$

so

$$\text{Var}_\theta(T) \geq \frac{1}{nl_1(\theta)}.$$

This gives the familiar $1/n$ variance scale.

When Is the Bound Attained?

The only inequality in the proof was Cauchy–Schwarz.

Equality holds exactly when

$$T(X) - \mathbb{E}_\theta[T(X)] = c(\theta)S_\theta(X) \quad \text{almost surely.}$$

Since T is unbiased for $g(\theta)$,

$$T(X) - g(\theta) = c(\theta)S_\theta(X).$$

Using

$$\text{Cov}_\theta(T, S_\theta) = g'(\theta),$$

we obtain

$$c(\theta) = \frac{g'(\theta)}{I(\theta)}.$$

Thus equality requires

$$T(X) - g(\theta) = \frac{g'(\theta)}{I(\theta)} S_\theta(X). \quad \text{(very restrictive condition.)}$$

Example: Gaussian Mean

Suppose

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2), \quad \sigma^2 \text{ known.}$$

For one observation,

$$I_1(\mu) = \frac{1}{\sigma^2} \implies I_n(\mu) = \frac{n}{\sigma^2}.$$

Any unbiased estimator of μ therefore satisfies

$$\text{Var}_\mu(T) \geq \frac{\sigma^2}{n}.$$

But

$$\mathbb{E}_\mu[\bar{X}] = \mu, \quad \text{Var}_\mu(\bar{X}) = \frac{\sigma^2}{n}.$$

Hence

$\bar{X}$ attains the Cramér–Rao bound.

Example: Bernoulli Model

Suppose

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta).$$

For one observation,

$$\ell(\theta; X) = X \log \theta + (1 - X) \log(1 - \theta),$$

so

$$S_\theta(X) = \frac{X - \theta}{\theta(1 - \theta)}.$$

Therefore

$$l_1(\theta) = \frac{1}{\theta(1 - \theta)}, \quad l_n(\theta) = \frac{n}{\theta(1 - \theta)}.$$

Thus every unbiased estimator of θ satisfies

$$\text{Var}_\theta(T) \geq \frac{\theta(1 - \theta)}{n}.$$

The Sample Mean Is Efficient

For Bernoulli observations,

$$\hat{\theta} = \bar{X}$$

is unbiased and

$$\text{Var}_{\theta}(\bar{X}) = \frac{\theta(1 - \theta)}{n}.$$

This exactly matches the Cramér–Rao lower bound:

$$\boxed{\text{Var}_{\theta}(\bar{X}) = \frac{1}{I_n(\theta)}}.$$

An unbiased estimator attaining the Cramér–Rao lower bound is called **efficient**.

Thus $\bar{X}$ is an efficient estimator of θ in the Bernoulli model.

Efficiency

Suppose T is unbiased for $g(\theta)$. The Cramér–Rao bound gives

$$\mathrm{Var}_{\theta}(T) \geq \frac{[g'(\theta)]^2}{I(\theta)}.$$

Define the **efficiency** of T at θ by

$$\mathrm{Eff}_{\theta}(T) = \frac{[g'(\theta)]^2 / I(\theta)}{\mathrm{Var}_{\theta}(T)}.$$

Therefore

$$0 \leq \mathrm{Eff}_{\theta}(T) \leq 1.$$

If

$$\mathrm{Eff}_{\theta}(T) = 1,$$

then T attains the Cramér–Rao lower bound at θ .

Relative Efficiency

Sometimes we want to compare two estimators directly.

If T_1 and T_2 are unbiased for the same quantity, define

$$\text{RE}_\theta(T_1, T_2) = \frac{\text{Var}_\theta(T_2)}{\text{Var}_\theta(T_1)}.$$

If

$$\text{RE}_\theta(T_1, T_2) > 1,$$

then T_1 has smaller variance and is more efficient at θ .

Efficiency is therefore a way of comparing how effectively different estimators use the information available in the data.

Example: Mean versus Median

Suppose

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2).$$

The sample mean satisfies

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

For large n , the sample median $\tilde{X}$ satisfies

$$\text{Var}(\tilde{X}) \approx \frac{\pi\sigma^2}{2n}.$$

Thus the asymptotic relative efficiency of the median to the mean is

$$\text{ARE}(\tilde{X}, \bar{X}) = \frac{2}{\pi} \approx 0.637.$$

Under Gaussian sampling, the mean uses the available information more efficiently.

But Efficiency Is Model Dependent

The previous comparison assumed exactly

$$X_i \sim N(\mu, \sigma^2).$$

The sample mean is highly efficient under Gaussian sampling, but it can be sensitive to extreme observations.

The median sacrifices some Gaussian efficiency in exchange for robustness.

Thus

efficiency is always relative to a statistical model.

A procedure that is optimal under one model need not perform well when that model is misspecified.

A Limitation of the Cramér–Rao Bound

The classical Cramér–Rao inequality applies to **unbiased** estimators.

Suppose instead

$$\mathbb{E}_\theta[T] = m(\theta).$$

Applying the same covariance argument gives

$$\mathrm{Var}_\theta(T) \geq \frac{[m'(\theta)]^2}{I(\theta)}.$$

But the mean squared error is

$$\mathrm{MSE}_\theta(T) = \mathrm{Var}_\theta(T) + [m(\theta) - g(\theta)]^2.$$

Thus allowing bias changes the optimization problem.

An estimator can have variance below the unbiased Cramér–Rao bound.

Bias Can Improve Risk

Consider

$$X \sim N(\theta, 1)$$

and the shrinkage estimator

$$\delta_\lambda(X) = \lambda X, \quad 0 < \lambda < 1.$$

Its bias is

$$\mathbb{E}_\theta[\delta_\lambda(X)] - \theta = -(1 - \lambda)\theta,$$

while

$$\text{Var}_\theta(\delta_\lambda) = \lambda^2 < 1.$$

Its squared-error risk is

$$R(\theta, \delta_\lambda) = \lambda^2 + (1 - \lambda)^2 \theta^2.$$

Thus introducing bias can substantially reduce variance and risk for some parameter values.

Cramér–Rao versus Decision Theory

Cramér–Rao asks:

Among unbiased estimators, how small can variance be?

Decision theory asks:

Among all decision rules, how small can risk be?

These are different optimization problems.

The distinction is important:

unbiasedness may exclude useful shrinkage estimators.

James–Stein is an extreme example: allowing bias can produce an estimator with uniformly smaller squared-error risk in dimension $d \geq 3$.

When Can Cramér–Rao Be Attained?

Suppose $T(X)$ is unbiased for $g(\theta)$ and attains the Cramér–Rao bound for every θ .

From the equality condition,

$$T(X) - g(\theta) = \frac{g'(\theta)}{I(\theta)} S_{\theta}(X).$$

Equivalently,

$$S_{\theta}(X) = a(\theta) T(X) + b(\theta),$$

where

$$a(\theta) = \frac{I(\theta)}{g'(\theta)}, \quad b(\theta) = -a(\theta)g(\theta).$$

Thus the score must depend on the data only through $T(X)$.

Integrating the Score

Since

$$S_{\theta}(x) = \frac{\partial}{\partial \theta} \log f_{\theta}(x),$$

the equality condition gives

$$\frac{\partial}{\partial \theta} \log f_{\theta}(x) = a(\theta) T(x) + b(\theta).$$

Integrating with respect to θ ,

$$\log f_{\theta}(x) = \eta(\theta) T(x) - A(\theta) + C(x),$$

where

$$\eta'(\theta) = a(\theta), \quad -A'(\theta) = b(\theta).$$

Exponentiating,

$$f_{\theta}(x) = h(x) \exp\{\eta(\theta) T(x) - A(\theta)\}.$$

This is an exponential-family form.

Why Exponential Families Appear

The Cramér–Rao equality condition requires

$$S_{\theta}(X) = a(\theta)T(X) + b(\theta).$$

But this says that the derivative of the log-likelihood is affine in a statistic $T(X)$.

Integrating produces

$$f_{\theta}(x) = h(x) \exp\{\eta(\theta)T(x) - A(\theta)\}.$$

Hence exponential families are naturally connected to efficient estimation.

Exact attainment of Cramér–Rao $\implies$ strong exponential-family structure.

Canonical Exponential Family

Write a one-parameter exponential family in canonical form:

$$f_{\eta}(x) = h(x) \exp\{\eta T(x) - A(\eta)\}.$$

The function

$$A(\eta)$$

is the **log-partition function**.

Since the density integrates to one,

$$1 = \int h(x) e^{\eta T(x) - A(\eta)} dx,$$

so

$$A(\eta) = \log \int h(x) e^{\eta T(x)} dx.$$

The derivatives of A encode important properties of the sufficient statistic $T(X)$.

First Derivative of the Log-Partition Function

Differentiate

$$e^{A(\eta)} = \int h(x) e^{\eta T(x)} dx.$$

Then

$$A'(\eta) e^{A(\eta)} = \int T(x) h(x) e^{\eta T(x)} dx.$$

Dividing by $e^{A(\eta)}$ gives

$$A'(\eta) = \int T(x) f_{\eta}(x) dx.$$

Therefore

$$\boxed{A'(\eta) = \mathbb{E}_{\eta}[T(X)]}.$$

Thus the gradient of the log-partition function gives the mean of the canonical statistic.

Second Derivative of the Log-Partition Function

Differentiate again:

$$A''(\eta) = \frac{d}{d\eta} \mathbb{E}_\eta[T(X)].$$

Using the exponential-family density,

$$A''(\eta) = \mathbb{E}_\eta[T(X)^2] - \mathbb{E}_\eta[T(X)]^2.$$

Hence

$$A''(\eta) = \text{Var}_\eta(T(X)).$$

Therefore

$$A''(\eta) \geq 0,$$

so A is convex.

For a nondegenerate family,

$$A''(\eta) > 0,$$

and A is strictly convex.

Fisher Information in an Exponential Family

For

$$f_{\eta}(x) = h(x) \exp\{\eta T(x) - A(\eta)\},$$

the score is

$$S_{\eta}(X) = T(X) - A'(\eta).$$

Since

$$A'(\eta) = \mathbb{E}_{\eta}[T(X)],$$

we have

$$S_{\eta}(X) = T(X) - \mathbb{E}_{\eta}[T(X)].$$

Therefore

$$I(\eta) = \text{Var}_{\eta}(T(X)) = A''(\eta).$$

Thus

$$\boxed{I(\eta) = A''(\eta).}$$

Fisher information is exactly the curvature of the log-partition function.

Example: Bernoulli as an Exponential Family

For $X \sim \text{Bernoulli}(p)$,

$$f_p(x) = p^x(1-p)^{1-x}.$$

Define the natural parameter

$$\eta = \log \frac{p}{1-p}.$$

Then

$$p = \frac{e^\eta}{1 + e^\eta},$$

and

$$f_\eta(x) = \exp \{ \eta x - \log(1 + e^\eta) \}.$$

Thus

$$T(x) = x, \quad A(\eta) = \log(1 + e^\eta).$$

Hence

$$A'(\eta) = p, \quad A''(\eta) = p(1-p) \implies I(\eta) = p(1-p).$$

Fisher Information Depends on Parameterization

Suppose the model is parameterized by θ , and define a new parameter

$$\eta = \eta(\theta).$$

The score with respect to θ is

$$S_{\theta}(X) = \frac{\partial}{\partial \theta} \log f_{\eta(\theta)}(X) = \eta'(\theta) S_{\eta}(X).$$

Therefore

$$I_{\theta}(\theta) = \mathbb{E}_{\theta}[S_{\theta}(X)^2] = [\eta'(\theta)]^2 I_{\eta}(\eta(\theta)).$$

Hence

$$I_{\theta}(\theta) = I_{\eta}(\eta(\theta)) \left(\frac{d\eta}{d\theta} \right)^2.$$

Fisher information transforms with the square of the change of scale.

Example: Bernoulli Revisited

For the natural parameter

$$\eta = \log \frac{p}{1-p},$$

we found

$$I_{\eta}(\eta) = p(1-p).$$

Since

$$\frac{d\eta}{dp} = \frac{1}{p(1-p)},$$

the information for p is

$$I_p(p) = I_{\eta}(\eta) \left(\frac{d\eta}{dp} \right)^2.$$

Therefore

$$I_p(p) = p(1-p) \frac{1}{p^2(1-p)^2} = \boxed{\frac{1}{p(1-p)}}.$$

The two formulas describe the same model under different coordinates.

KL Divergence Between Nearby Models

Consider two nearby parameter values

$$\theta \quad \text{and} \quad \theta + \Delta.$$

Their KL divergence is

$$D_{\text{KL}}(P_{\theta} \| P_{\theta+\Delta}) = \mathbb{E}_{\theta} \left[\log \frac{f_{\theta}(X)}{f_{\theta+\Delta}(X)} \right].$$

What happens when Δ is small?

Taylor expand the log-likelihood around θ :

$$\log f_{\theta+\Delta}(X) = \log f_{\theta}(X) + \Delta S_{\theta}(X) + \frac{\Delta^2}{2} \partial_{\theta}^2 \log f_{\theta}(X) + o(\Delta^2).$$

Local Expansion of KL Divergence

Substituting the Taylor expansion,

$$D_{\text{KL}}(P_\theta \| P_{\theta+\Delta}) = -\Delta \mathbb{E}_\theta[S_\theta(X)] - \frac{\Delta^2}{2} \mathbb{E}_\theta[\partial_\theta^2 \log f_\theta(X)] + o(\Delta^2).$$

Recall

$$\mathbb{E}_\theta[S_\theta(X)] = 0$$

and

$$-\mathbb{E}_\theta[\partial_\theta^2 \log f_\theta(X)] = I(\theta).$$

Therefore

$$D_{\text{KL}}(P_\theta \| P_{\theta+\Delta}) = \frac{1}{2} I(\theta) \Delta^2 + o(\Delta^2).$$

What Does Fisher Information Measure?

The local KL expansion gives

$$D_{\text{KL}}(P_{\theta} \| P_{\theta+\Delta}) \approx \frac{1}{2} I(\theta) \Delta^2.$$

Thus Fisher information measures how rapidly the distribution changes as the parameter changes.

If $I(\theta)$ is large,

$$P_{\theta} \quad \text{and} \quad P_{\theta+\Delta}$$

become distinguishable even for small Δ .

If $I(\theta)$ is small, nearby parameter values generate very similar distributions.

Fisher information measures local statistical distinguishability.

Why the First-Order Term Vanishes

KL divergence satisfies

$$D_{\text{KL}}(P_{\theta} \| P_{\vartheta}) \geq 0$$

with equality at

$$\vartheta = \theta.$$

Thus, as a function of ϑ , KL divergence has a local minimum at $\vartheta = \theta$.

Consequently,

$$\left. \frac{\partial}{\partial \vartheta} D_{\text{KL}}(P_{\theta} \| P_{\vartheta}) \right|_{\vartheta=\theta} = 0.$$

Its first nonzero local term is therefore quadratic:

$$D_{\text{KL}}(P_{\theta} \| P_{\theta+\Delta}) \approx \frac{1}{2} I(\theta) \Delta^2.$$

So Fisher information plays the role of a local **curvature**.

Example: Gaussian Location

Let

$$P_\mu = N(\mu, \sigma^2).$$

For two means μ and $\mu + \Delta$,

$$D_{\text{KL}}(N(\mu, \sigma^2) \| N(\mu + \Delta, \sigma^2)) = \frac{\Delta^2}{2\sigma^2}.$$

But

$$I(\mu) = \frac{1}{\sigma^2}.$$

Hence

$$D_{\text{KL}}(P_\mu \| P_{\mu+\Delta}) = \frac{1}{2} I(\mu) \Delta^2.$$

In this model the quadratic approximation is actually exact.

Information as Local Geometry

For nearby parameters,

$$D_{\text{KL}}(P_\theta \| P_{\theta+d\theta}) \approx \frac{1}{2} I(\theta) (d\theta)^2.$$

This suggests viewing

$$I(\theta)(d\theta)^2$$

as a local squared distance between nearby statistical models.

Under a reparameterization $\eta = \eta(\theta)$,

$$I_\theta(\theta)(d\theta)^2 = I_\eta(\eta)(d\eta)^2.$$

Thus, although the numerical value of Fisher information changes with coordinates, the local quantity

$$I(\theta)(d\theta)^2$$

does not.

This is the beginning of **information geometry**.

Multivariate Parameters

Now suppose

$$\theta = (\theta_1, \dots, \theta_d)^\top \in \mathbb{R}^d.$$

The score becomes a vector:

$$S_\theta(X) = \nabla_\theta \log f_\theta(X).$$

Under regularity conditions,

$$\mathbb{E}_\theta[S_\theta(X)] = 0.$$

The **Fisher information matrix** is

$$I(\theta) = \mathbb{E}_\theta [S_\theta(X) S_\theta(X)^\top].$$

Equivalently,

$$I(\theta) = -\mathbb{E}_\theta [\nabla_\theta^2 \log f_\theta(X)].$$

Multivariate Cramér–Rao Bound

Suppose $T(X) \in \mathbb{R}^k$ is unbiased for

$$g(\theta) \in \mathbb{R}^k.$$

Let

$$G(\theta) = \nabla_{\theta} g(\theta)$$

be the $k \times d$ Jacobian.

Then, under regularity conditions,

$$\text{Cov}_{\theta}(T) \succeq G(\theta)I(\theta)^{-1}G(\theta)^{\top}.$$

Here $A \succeq B$ means

$$A - B$$

is positive semidefinite.

For estimating θ itself,

$$g(\theta) = \theta, \quad G(\theta) = I_d,$$

so

Local KL Geometry in Multiple Dimensions

Let $\Delta \in \mathbb{R}^d$ be small.

A second-order Taylor expansion gives

$$D_{\text{KL}}(P_\theta \| P_{\theta+\Delta}) = -\Delta^\top \mathbb{E}_\theta[S_\theta(X)] - \frac{1}{2} \Delta^\top \mathbb{E}_\theta[\nabla_\theta^2 \log f_\theta(X)] \Delta + o(\|\Delta\|^2).$$

Using

$$\mathbb{E}_\theta[S_\theta(X)] = 0$$

and

$$I(\theta) = -\mathbb{E}_\theta[\nabla_\theta^2 \log f_\theta(X)],$$

we obtain

$$D_{\text{KL}}(P_\theta \| P_{\theta+\Delta}) = \frac{1}{2} \Delta^\top I(\theta) \Delta + o(\|\Delta\|^2).$$

Fisher Information Defines Local Geometry

The quantity

$$\Delta^\top I(\theta) \Delta$$

measures how distinguishable P_θ and $P_{\theta+\Delta}$ are locally.

Different directions can contain different amounts of information.

If

$$\Delta^\top I(\theta) \Delta$$

is large, moving in direction Δ changes the distribution substantially.

If it is small, the model changes very little in that direction.

Thus $I(\theta)$ acts as a local metric on the parameter space:

$$ds^2 = d\theta^\top I(\theta) d\theta.$$

This is the **Fisher information metric**.

Lecture 8: Takeaways

- Unbiasedness requires $\mathbb{E}_\theta[T] = g(\theta)$ for every θ .
- The **score** measures local sensitivity of the likelihood.
- Its variance is the **Fisher information**.
- Cramér–Rao gives

$$\mathrm{Var}_\theta(T) \geq \frac{[g'(\theta)]^2}{I(\theta)}$$

for unbiased estimators.

- Attaining Cramér–Rao is closely connected to exponential-family structure.
- In a canonical exponential family,

$$A'(\eta) = \mathbb{E}_\eta[T], \quad A''(\eta) = \mathrm{Var}_\eta(T) = I(\eta).$$

- Locally,

$$D_{\mathrm{KL}}(P_\theta \| P_{\theta+\Delta}) \approx \frac{1}{2} \Delta^\top I(\theta) \Delta.$$

Fisher information connects estimation, likelihood curvature, and statistical geometry.